

**PENETRATION
TESTING AND VULNERABILITY
ANALYSIS LAB**

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Assessment – III

ADVANCED SERVICE ENUMERATION

1) `nmap -p 80,443,8080 --script http-title,http-server-header,http-methods 98.82.161.185`

1. Target

- 98.82.161.185 → the host you're scanning

2. Ports

- `-p 80,443,8080`
 - 80 → HTTP
 - 443 → HTTPS
 - 8080 → Common alternate HTTP / proxy / app port

You're limiting the scan to web-related ports only (good OPSEC move).

3. NSE scripts used

You're running three HTTP NSE scripts:

`http-title`

- Fetches the `<title>` tag from the webpage
- Useful for:
 - Identifying apps (e.g., "Apache Tomcat", "Welcome to nginx")
 - Spotting admin panels or dashboards

`http-server-header`

- Grabs the `Server :` HTTP response header
- Often reveals:
 - Web server type (Apache, nginx, IIS)
 - Sometimes version info (if misconfigured)

`http-methods`

- Checks which HTTP methods are allowed
- Looks for risky ones like:
 - PUT
 - DELETE
 - TRACE
 - CONNECT

This is especially useful for spotting misconfigurations.

```
$ nmap -p 80,443,8080 --script http-title,http-server-header,http-methods 98.82.161.185
Starting Nmap 7.92 ( https://nmap.org ) at 2026-02-06 14:48 IST
Nmap scan report for ec2-98-82-161-185.compute-1.amazonaws.com (98.82.161.185)
Host is up (0.29s latency).

PORT      STATE      SERVICE
80/tcp    open      http
|_http-server-header: Server
|_http-title: Did not follow redirect to https://ec2-98-82-161-185.compute-1.amazonaws.com/
|_http-methods:
|_ Supported Methods: GET HEAD POST OPTIONS
443/tcp    open      https
|_http-title: 400 The plain HTTP request was sent to HTTPS port
|_http-server-header: Server
8080/tcp   filtered  http-proxy

Nmap done: 1 IP address (1 host up) scanned in 14.69 seconds
```

2) `nmap -p 80,443 --script http-headers,http-title,http-server-header 98.82.161.185`

- Scans **port 80 (HTTP)** only on the target.
- Checks **allowed HTTP methods** specifically on /up loads.
- Looks for dangerous methods like **PUT / DELETE** (file upload or removal risk).
- Helps detect **misconfigured upload directories**

```

14:50:14 ○ [krxsna@pavilion] system~
$ nmap -p 80,443 --script http-headers,http-title,http-server-header 98.82.161.185
Starting Nmap 7.92 ( https://nmap.org ) at 2026-02-06 14:50 IST
Nmap scan report for ec2-98-82-161-185.compute-1.amazonaws.com (98.82.161.185)
Host is up (0.33s latency).

PORT      STATE SERVICE
80/tcp    open  http
|_http-title: Did not follow redirect to https://ec2-98-82-161-185.compute-1.amazonaws.com/
|_http-headers:
|   Server: Server
|   Date: Fri, 06 Feb 2026 09:20:21 GMT
|   Content-Type: text/html
|   Content-Length: 163
|   Connection: close
|   Location: https://ec2-98-82-161-185.compute-1.amazonaws.com/
|_
|_ (Request type: GET)
|_http-server-header: Server
443/tcp    open  https
|_http-headers:
|   Server: Server
|   Date: Fri, 06 Feb 2026 09:20:25 GMT
|   Content-Type: text/html
|   Transfer-Encoding: chunked
|   Connection: close
|_
|_ (Request type: GET)
|_http-title: x
|_http-server-header: Server

Nmap done: 1 IP address (1 host up) scanned in 7.32 seconds

```

•

3) `nmap -p 80,443 --script http-enum 98.82.161.185`

Scans **HTTPS (port 443 and 80)** to enumerate supported **SSL/TLS ciphers**.

- Identifies **weak or deprecated protocols** (SSLv3, TLS 1.0/1.1).
- `-Pn` skips host discovery and **forces the scan** even if ping is blocked.
- Helps assess **TLS security posture** and compliance issues.

```
$ nmap -p 80,443 --script http-enum 98.82.161.185
Starting Nmap 7.92 ( https://nmap.org ) at 2026-02-06 14:50 IST
Nmap scan report for ec2-98-82-161-185.compute-1.amazonaws.com (98.82.161.185)
Host is up (0.33s latency).

PORT      STATE SERVICE
80/tcp    open  http
443/tcp   open  https

Nmap done: 1 IP address (1 host up) scanned in 122.78 seconds
```

4) `nmap -p 80,443 --script http-generator,http-waf-detect 98.82.161.185`
Scans **HTTP and HTTPS** services on ports **80 and 443**.

1. `http-generator` identifies the **backend technology/framework** (CMS, app stack).
2. `http-waf-detect` checks for presence of a **Web Application Firewall**.
3. Useful for **fingerprinting defenses** before deeper web testing.

```
➤ nmap -p 80,443 --script http-generator,http-waf-detect 98.82.161.185
Starting Nmap 7.92 ( https://nmap.org ) at 2026-02-06 14:51 IST
Nmap scan report for ec2-98-82-161-185.compute-1.amazonaws.com (98.82.161.185)
Host is up (0.51s latency).

PORT      STATE SERVICE
80/tcp    open  http
443/tcp   open  https

Nmap done: 1 IP address (1 host up) scanned in 24.25 seconds
```

5) `nmap -p 80 --script http-methods --script-args http-methods.url-path='/uploads'`
`98.82.161.185`

Scans **HTTP port 80** on the target.

- Tests **allowed HTTP methods** specifically on `/uploads`.
- Flags dangerous methods like **PUT/DELETE** (upload or file-tampering risk).
- Helps detect **insecure upload directory configurations**.

```

14.02.02 ~ [kix@redpotion] ~$ nmap -p 80 --script http-methods --script-args http-methods.url-path='/uploads' 98.82.161.185
Starting Nmap 7.92 ( https://nmap.org ) at 2026-02-06 14:54 IST
Nmap scan report for ec2-98-82-161-185.compute-1.amazonaws.com (98.82.161.185)
Host is up (0.31s latency).

PORT      STATE SERVICE
80/tcp    open  http
| http-methods:
|   Supported Methods: GET HEAD POST OPTIONS
|_  Path tested: /uploads

Nmap done: 1 IP address (1 host up) scanned in 5.73 seconds

```

6) `nmap -p 443 --script ssl-heartbleed,ssl-poodle,ssl-dh-params 98.82.161.185`
 Scans **HTTPS (port 443)** for **critical SSL/TLS vulnerabilities**.

- `ssl-heartbleed` checks exposure to the **Heartbleed** bug.
- `ssl-poodle` tests for **POODLE** (SSLv3 downgrade weakness).
- `ssl-dh-params` analyzes **Diffie-Hellman strength** (weak primes / Logjam risk).

```

$ nmap -p 443 --script ssl-heartbleed,ssl-poodle,ssl-dh-params 98.82.161.185
Starting Nmap 7.92 ( https://nmap.org ) at 2026-02-06 14:55 IST
Nmap scan report for 98.82.161.185
Host is up (0.44s latency).

PORT      STATE SERVICE
443/tcp    open  https

Nmap done: 1 IP address (1 host up) scanned in 36.69 seconds

```

7) `nmap -p 443 --script ssl-enum-ciphers -Pn 98.82.161.185`

Scans **HTTPS on port 443** to list supported **SSL/TLS cipher suites**.

1. Reveals **weak or deprecated ciphers/protocols** (e.g., TLS 1.0/1.1).
2. `-Pn` skips host discovery and **forces the scan** even if ICMP is blocked.
3. Useful for **TLS hardening and compliance checks**.

```
$ nmap -p 443 --script ssl-enum-ciphers -Pn 98.82.161.185
Starting Nmap 7.92 ( https://nmap.org ) at 2026-02-06 14:58 IST
Nmap scan report for ec2-98-82-161-185.compute-1.amazonaws.com (98.82.161.185)
Host is up (0.28s latency).

PORT      STATE SERVICE
443/tcp   open  https
| ssl-enum-ciphers:
|   TLSv1.0:
|     ciphers:
|       TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA (ecdh_x25519) - A
|       TLS_RSA_WITH_AES_128_CBC_SHA (rsa 2048) - A
|     compressors:
|       NULL
|     cipher preference: server
|   TLSv1.1:
|     ciphers:
|       TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA (ecdh_x25519) - A
|       TLS_RSA_WITH_AES_128_CBC_SHA (rsa 2048) - A
|     compressors:
|       NULL
|     cipher preference: server
|   TLSv1.2:
|     ciphers:
|       TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256 (ecdh_x25519) - A
|       TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA256 (ecdh_x25519) - A
|       TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA (ecdh_x25519) - A
|       TLS_RSA_WITH_AES_128_GCM_SHA256 (rsa 2048) - A
|       TLS_RSA_WITH_AES_128_CBC_SHA256 (rsa 2048) - A
|       TLS_RSA_WITH_AES_128_CBC_SHA (rsa 2048) - A
|     compressors:
|       NULL
|     cipher preference: server
|   TLSv1.3:
|     ciphers:
|       TLS_AKE_WITH_AES_128_GCM_SHA256 (ecdh_x25519) - A
|       TLS_AKE_WITH_AES_256_GCM_SHA384 (ecdh_x25519) - A
|     cipher preference: server
|_  least strength: A

Nmap done: 1 IP address (1 host up) scanned in 16.62 seconds
```

8) `nmap -p 443 --script ssl-cert-intaddr,ssl-known-key 98.82.161.185`

Scans **HTTPS on port 443** for certificate weaknesses.

- `ssl-cert-intaddr` detects **internal IPs/hostnames** leaked in the cert.
- `ssl-known-key` checks for **known or compromised private keys**.
- Useful for spotting **info leakage and crypto misconfigurations**.

```
$ nmap -p 443 --script ssl-enum-ciphers -Pn 98.82.161.185
Starting Nmap 7.92 ( https://nmap.org ) at 2026-02-06 14:58 IST
Nmap scan report for ec2-98-82-161-185.compute-1.amazonaws.com (98.82.161.185)
Host is up (0.28s latency).

PORT      STATE SERVICE
443/tcp   open  https
| ssl-enum-ciphers:
|   TLSv1.0:
|     ciphers:
|       TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA (ecdh_x25519) - A
|       TLS_RSA_WITH_AES_128_CBC_SHA (rsa 2048) - A
|     compressors:
|       NULL
|     cipher preference: server
|   TLSv1.1:
|     ciphers:
|       TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA (ecdh_x25519) - A
|       TLS_RSA_WITH_AES_128_CBC_SHA (rsa 2048) - A
|     compressors:
|       NULL
|     cipher preference: server
|   TLSv1.2:
|     ciphers:
|       TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256 (ecdh_x25519) - A
|       TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA256 (ecdh_x25519) - A
|       TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA (ecdh_x25519) - A
|       TLS_RSA_WITH_AES_128_GCM_SHA256 (rsa 2048) - A
|       TLS_RSA_WITH_AES_128_CBC_SHA256 (rsa 2048) - A
|       TLS_RSA_WITH_AES_128_CBC_SHA (rsa 2048) - A
|     compressors:
|       NULL
|     cipher preference: server
|   TLSv1.3:
|     ciphers:
|       TLS_AKE_WITH_AES_128_GCM_SHA256 (ecdh_x25519) - A
|       TLS_AKE_WITH_AES_256_GCM_SHA384 (ecdh_x25519) - A
|     cipher preference: server
|_  least strength: A

Nmap done: 1 IP address (1 host up) scanned in 16.62 seconds

14:58:45 ○ [krxsna@pavilion] system~
$ nmap -p 443 --script ssl-cert-intaddr,ssl-known-key 98.82.161.185
Starting Nmap 7.92 ( https://nmap.org ) at 2026-02-06 15:10 IST
Nmap scan report for ec2-98-82-161-185.compute-1.amazonaws.com (98.82.161.185)
Host is up (0.29s latency).

PORT      STATE SERVICE
443/tcp   open  https

Nmap done: 1 IP address (1 host up) scanned in 1.52 seconds
```


9) nmap -p 443 --script ssl-enum-ciphers --script-args sslciphers=all 98.82.161.185
Scans **HTTPS (port 443)** for **all supported SSL/TLS cipher suites**.

1. Forces enumeration of **legacy and modern ciphers** (sslciphers=all).
2. Helps identify **weak, deprecated, or insecure ciphers** still enabled.
3. Useful for **deep TLS hardening and audit** checks.

```
Starting Nmap 7.92 ( https://nmap.org ) at 2026-02-06 15:10 IST
Nmap scan report for ec2-98-82-161-185.compute-1.amazonaws.com (98.82.161.185)
Host is up (0.29s latency).

PORT      STATE SERVICE
443/tcp   open  https

Nmap done: 1 IP address (1 host up) scanned in 1.52 seconds

15:10:17 ○ [krxsna@pavilion] system~
$ nmap -p 443 --script ssl-enum-ciphers --script-args sslciphers=all 98.82.161.185
Starting Nmap 7.92 ( https://nmap.org ) at 2026-02-06 15:10 IST
Nmap scan report for ec2-98-82-161-185.compute-1.amazonaws.com (98.82.161.185)
Host is up (0.29s latency).

PORT      STATE SERVICE
443/tcp   open  https
| ssl-enum-ciphers:
|   TLSv1.0:
|   | ciphers:
|   |   TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA (ecdh_x25519) - A
|   |   TLS_RSA_WITH_AES_128_CBC_SHA (rsa 2048) - A
|   | compressors:
|   |   NULL
|   | cipher preference: server
|   TLSv1.1:
|   | ciphers:
|   |   TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA (ecdh_x25519) - A
|   |   TLS_RSA_WITH_AES_128_CBC_SHA (rsa 2048) - A
|   | compressors:
|   |   NULL
|   | cipher preference: server
|   TLSv1.2:
|   | ciphers:
|   |   TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256 (ecdh_x25519) - A
|   |   TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA256 (ecdh_x25519) - A
|   |   TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA (ecdh_x25519) - A
|   |   TLS_RSA_WITH_AES_128_GCM_SHA256 (rsa 2048) - A
|   |   TLS_RSA_WITH_AES_128_CBC_SHA256 (rsa 2048) - A
|   |   TLS_RSA_WITH_AES_128_CBC_SHA (rsa 2048) - A
|   | compressors:
|   |   NULL
|   | cipher preference: server
|   TLSv1.3:
|   | ciphers:
|   |   TLS_AKE_WITH_AES_128_GCM_SHA256 (ecdh_x25519) - A
|   |   TLS_AKE_WITH_AES_256_GCM_SHA384 (ecdh_x25519) - A
|   | cipher preference: server
|   |_ least strength: A

Nmap done: 1 IP address (1 host up) scanned in 13.84 seconds
```

10) `nmap -p 445 --script smb-os-discovery,smb2-security-mode,smb2-time 98.82.161.185`

Key points (3-4):

1. Scans **SMB service on port 445**.
2. `smb-os-discovery` attempts to identify the **remote OS and hostname**.
3. `smb2-security-mode` checks **SMB signing and security settings**.
4. `smb2-time` retrieves the server's **system time** (useful for skew/auth issues).

```

Starting Nmap 7.92 ( https://nmap.org ) at 2026-02-06 15:10 IST
Nmap scan report for ec2-98-82-161-185.compute-1.amazonaws.com (98.82.161.185)
Host is up (0.29s latency).

PORT      STATE SERVICE
443/tcp   open  https
|_ ssl-enum-ciphers:
|   TLSv1.0:
|     ciphers:
|       TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA (ecdh_x25519) - A
|       TLS_RSA_WITH_AES_128_CBC_SHA (rsa 2048) - A
|     compressors:
|       NULL
|     cipher preference: server
|   TLSv1.1:
|     ciphers:
|       TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA (ecdh_x25519) - A
|       TLS_RSA_WITH_AES_128_CBC_SHA (rsa 2048) - A
|     compressors:
|       NULL
|     cipher preference: server
|   TLSv1.2:
|     ciphers:
|       TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256 (ecdh_x25519) - A
|       TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA256 (ecdh_x25519) - A
|       TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA (ecdh_x25519) - A
|       TLS_RSA_WITH_AES_128_GCM_SHA256 (rsa 2048) - A
|       TLS_RSA_WITH_AES_128_CBC_SHA256 (rsa 2048) - A
|       TLS_RSA_WITH_AES_128_CBC_SHA (rsa 2048) - A
|     compressors:
|       NULL
|     cipher preference: server
|   TLSv1.3:
|     ciphers:
|       TLS_AKE_WITH_AES_128_GCM_SHA256 (ecdh_x25519) - A
|       TLS_AKE_WITH_AES_256_GCM_SHA384 (ecdh_x25519) - A
|     cipher preference: server
|_ least strength: A

Nmap done: 1 IP address (1 host up) scanned in 13.84 seconds

15:11:13 ○ [krxsna@pavilion] system~
$ nmap -p 445 --script smb-os-discovery,smb2-security-mode,smb2-time 98.82.161.185
Starting Nmap 7.92 ( https://nmap.org ) at 2026-02-06 15:13 IST
Nmap scan report for ec2-98-82-161-185.compute-1.amazonaws.com (98.82.161.185)
Host is up (0.28s latency).

PORT      STATE SERVICE
445/tcp   filtered microsoft-ds

Nmap done: 1 IP address (1 host up) scanned in 3.42 seconds

```

11) nmap -p 445 --script smb-enum-shares,smb-enum-users 98.82.161.185
 Scans **SMB on port 445**.

- smb-enum-shares lists **accessible SMB shares** (including anonymous ones).
- smb-enum-users attempts to enumerate **user accounts** via SMB.
- Useful for spotting **information exposure and weak SMB configs**.

```

$ nmap -p 445 --script smb-os-discovery,smb2-security-mode,smb2-time 98.82.161.185
Starting Nmap 7.92 ( https://nmap.org ) at 2026-02-06 15:13 IST
Nmap scan report for ec2-98-82-161-185.compute-1.amazonaws.com (98.82.161.185)
Host is up (0.28s latency).

PORT      STATE      SERVICE
445/tcp    filtered  microsoft-ds

Nmap done: 1 IP address (1 host up) scanned in 3.42 seconds

15:13:44 ○ [krxsna@pavilion] system~
$ nmap -p 445 --script smb-enum-shares,smb-enum-users 98.82.161.185
Starting Nmap 7.92 ( https://nmap.org ) at 2026-02-06 15:15 IST
Nmap scan report for ec2-98-82-161-185.compute-1.amazonaws.com (98.82.161.185)
Host is up (0.28s latency).

PORT      STATE      SERVICE
445/tcp    filtered  microsoft-ds

Nmap done: 1 IP address (1 host up) scanned in 3.49 seconds

```

12) nmap -p 445 --script smb-enum-shares,smb-enum-users 98.82.161.185

```

$ nmap -p 445 --script smb-os-discovery,smb2-security-mode,smb2-time 98.82.161.185
Starting Nmap 7.92 ( https://nmap.org ) at 2026-02-06 15:13 IST
Nmap scan report for ec2-98-82-161-185.compute-1.amazonaws.com (98.82.161.185)
Host is up (0.28s latency).

PORT      STATE      SERVICE
445/tcp    filtered  microsoft-ds

Nmap done: 1 IP address (1 host up) scanned in 3.42 seconds

15:13:44 ○ [krxsna@pavilion] system~
$ nmap -p 445 --script smb-enum-shares,smb-enum-users 98.82.161.185
Starting Nmap 7.92 ( https://nmap.org ) at 2026-02-06 15:15 IST
Nmap scan report for ec2-98-82-161-185.compute-1.amazonaws.com (98.82.161.185)
Host is up (0.28s latency).

PORT      STATE      SERVICE
445/tcp    filtered  microsoft-ds

Nmap done: 1 IP address (1 host up) scanned in 3.49 seconds

```

13) nmap -p 445 --script smb-vuln* 98.82.161.185
 Scans **SMB service on port 445**.

- Enumerates **shared folders** and their access permissions.
- Attempts to list **valid SMB user accounts**.
- Helps identify **unauthenticated access or information leakage**.

```
$ nmap -p 445 --script smb-vuln* 98.82.161.185
Starting Nmap 7.92 ( https://nmap.org ) at 2026-02-06 15:18 IST
Nmap scan report for ec2-98-82-161-185.compute-1.amazonaws.com (98.82.161.185)
Host is up (0.28s latency).

PORT      STATE      SERVICE
445/tcp    filtered  microsoft-ds

Nmap done: 1 IP address (1 host up) scanned in 3.50 seconds
```

14) `nmap -p 445 --script smb-protocols,smb-security-mode 98.82.161.185`

Scans **SMB on port 445**.

1. `smb-protocols` identifies **enabled SMB versions** (SMBv1/2/3).
2. `smb-security-mode` checks **authentication and signing requirements**.
3. Helps detect **legacy SMB usage and weak security settings**.

```
$ nmap -p 445 --script smb-protocols,smb-security-mode 98.82.161.185
Starting Nmap 7.92 ( https://nmap.org ) at 2026-02-06 15:19 IST
Nmap scan report for ec2-98-82-161-185.compute-1.amazonaws.com (98.82.161.185)
Host is up (0.29s latency).

PORT      STATE      SERVICE
445/tcp    filtered  microsoft-ds

Nmap done: 1 IP address (1 host up) scanned in 3.45 seconds
```

15) `nmap -sV -p 80,443,445 --script vuln 98.82.161.185`

```
$ nmap -sV -p 80,443,445 --script vuln 98.82.161.185
Starting Nmap 7.92 ( https://nmap.org ) at 2026-02-06 15:23 IST
Pre-scan script results:
| broadcast-avahi-dos:
|   Discovered hosts:
|     224.0.0.251
|   After NULL UDP avahi packet DoS (CVE-2011-1002).
|_  Hosts are all up (not vulnerable).
```

Scans **ports 80, 443, and 445** for web and SMB services.

1. `-sV` enables **service and version detection**.
2. `--script vuln` runs **known vulnerability checks** against detected services.

3. Useful for finding **publicly known CVEs and misconfigurations** quickly .

16) nmap --script vuln -oA output_name 98.82.161.185

```
nmap --script vuln -oA output_name 98.82.161.185
Starting Nmap 7.92 ( https://nmap.org ) at 2026-02-06 15:23 IST
Pre-scan script results:
| broadcast-avahi-dos:
|   Discovered hosts:
|     224.0.0.251
|   After NULL UDP avahi packet DoS (CVE-2011-1002).
|_  Hosts are all up (not vulnerable).
```

Runs **vulnerability-detection NSE scripts** against the target.

- Scans **default ports** (since none are specified).
- -oA output_name saves results in **all formats** (.nmap, .xml, .gnmap).
- Useful for **reporting and later analysis** of discovered issues.

17) nmap --script vuln -T4 98.82.161.185

```
$ nmap --script vuln -T4 98.82.161.185
Starting Nmap 7.92 ( https://nmap.org ) at 2026-02-06 15:25 IST
Pre-scan script results:
| broadcast-avahi-dos:
|   Discovered hosts:
|     224.0.0.251
|   After NULL UDP avahi packet DoS (CVE-2011-1002).
|_  Hosts are all up (not vulnerable).
```

Runs **vulnerability NSE scripts** against the target.

- Uses **default ports** and detected services.
- -T4 increases scan **speed/aggressiveness** (faster, slightly noisier).
- Good for **quick vulnerability assessment** when time matters.